Gabe Flanagin

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Education

Western Washington University, Bellingham, WA 2025–2027 (expected)
M.S. in Mathematics | GPA 4.00

Selected coursework: Measure Theory (Real Analysis), Functional Analysis, Fourier Analysis (Topics in Analysis), Fourier Series and PDEs, Numerical Linear Algebra, Numerical Analysis, Nonlinear Optimization, Probability, Mathematical Biology, AI for STEM, Robotics (in progress)

Western Washington University, Bellingham, WA 2021–2025
B.S. in Mathematics; Minor in Chemistry

Research Interests

Scientific machine learning for PDEs (neural operators, physics-informed methods); numerical PDEs and boundary integral methods; PDE-constrained optimization and optimal control; inverse problems; scientific and high-performance computing

Publications

Submitted

Flanagin, G., Escoto, A., and Chao, Z. (2026). "A Spatial Model of Chronic Wasting Disease and the Optimal Placement of a Sight-Based Cull." Submitted to *Bulletin of Mathematical Biology*. Code: github.com/gflanagin/CWD_codebase.

Published

Flanagin, G. and Berger, R. (2026). "Rationalizing Patterns in the Cation Ordering, Geometric Distortion, and Electronic Properties of a Class of I–V–VI_2 Chalcogenide Semiconductors." *Journal of Physical Chemistry C*. doi:10.1021/acs.jpcc.5c05853.

Research Experience

Scientific Machine Learning for Cryo-Electron Tomography Jun.–Aug. 2026
G-RIPS, Zuse Institute Berlin (ZIB) — Mentors: Noushin Hajarolasvadi and Frosina Stojanovska

- Fine-tuned the CryoSiam deep-learning framework to segment biological structures in 3D cryo-electron tomograms with low signal-to-noise and missing-wedge artifacts, developing PyTorch training pipelines within a large existing research codebase.
- Implemented and debugged a topology-preserving loss for thin, connected structures; identified a failure mode in which the loss collapsed to zero and corrected it.
- Ran GPU training experiments on HPC clusters using Linux, Slurm, and CUDA-enabled workflows, including debugging distributed multi-GPU training runs.

Spatial Modeling and Optimal Control of Chronic Wasting Disease 2026–Present
Lead author, with Alex Escoto and Dr. Zhen Chao

- Formulated a five-compartment (S–E–I–R–W) reaction–diffusion model of chronic wasting disease on terrain surfaces built from USGS elevation data, with land cover setting carrying capacity and slope setting an anisotropic diffusion tensor.
- Posed a PDE-constrained optimal control problem for a space- and time-dependent cull of clinically affected deer, and implemented the exact discrete adjoint of a surface finite element scheme in FEniCSx/PETSc, verified to second order by Taylor tests and optimized with L-BFGS-B on meshes of up to 370,000 nodes.
- Derived a closed-form bound on what such a cull can achieve, and showed that optimal effort forms a ring at the invasion's leading edge and that terrain and land cover cut the effort needed by 26–35%.

Boundary Element Methods for the Linearized Poisson–Boltzmann Equation 2026–Present
M.S. project — Advised by Dr. Zhen Chao

- Studying the second-kind boundary integral formulation of Juffer et al. for macromolecular electrostatics, including its hypersingular-kernel cancellation, solvability and conditioning.

- Implementing single- and double-layer potentials and verifying jump relations, transmission conditions, and far-field behavior.
- Implementing first- and second-kind formulations in 2D and 3D, validating on a disk and sphere, and comparing conditioning under refinement and accuracy as charges approach the interface.

Hamilton–Jacobi Equations and Max-Plus Approximation 2025
Senior project — Advised by Dr. Rick Barnard

- Implemented a max-plus method for approximating value functions governed by Hamilton–Jacobi–Bellman equations.
- Extended a quadratic under-approximation result from the C^2 setting to functions that are C^1 except at finitely many locally convex nonsmooth points.
- Generalized a semiconvexity-preservation theorem for the linear-quadratic regulator semigroup, proving preservation for every positive time step in a broader function class.

First-Principles Modeling of Chalcogenide Semiconductors 2024–2025
Advised by Dr. Robert Berger

- Used density-functional-theory calculations to study cation ordering and electronic properties in I–V–VI_2 chalcogenide semiconductors, and introduced a bonding-motif model that rationalizes their structural variation; first-author publication in the *Journal of Physical Chemistry C*.

Selected Projects

Machine-Learning-Assisted Optimization for 2D Polygon Packing Spring 2026
CSCI 597E course project, with D. Jackson, R. Rapoport, and W. Zuber

- Augmented a basin-hopping/L-BFGS packing solver with a PPO-trained warm-start policy and a transformer filter for pruning perturbations, trained on 274,000 optimization runs (135 million iterations) generated on an HTCondor cluster.
- Benchmarked both against best-known packings: the filter reached AUC ≈ 0.80 as a classifier but worsened results as a gate, and the warm-start matched random initialization.

Presentations and Talks

An Introduction to Spatiotemporal Rabies Models (with A. Escoto), WWU Math Colloquium Apr. 2026
Using Max-Plus Algebra to Compute Reachability in Dynamic Systems, WWU Scholars Week May 2025

Teaching Experience

Western Washington University 2025–Present
Graduate Teaching Assistant, Department of Mathematics

- Instructor — MATH 114: Precalculus I Fall 2026
- Instructor — MATH 115: Precalculus II Spring 2026
- TA Liaison — MATH 114: Precalculus I Winter 2026
- Instructor — MATH 112: Functions and Algebraic Methods Fall 2025

Awards and Honors

Richard Greene Scholarship, Western Washington University 2026

Programming and Computational Tools

Machine learning and HPC: Python, PyTorch, CUDA, Slurm, HTCondor, Linux, Bash, Git
Scientific computing: FEniCSx, PETSc, NumPy, SciPy, MATLAB, mmg, ParaView, VASP, Amira, QGIS